

Challenge Context

Good Nature Agro is a Zambian social enterprise that works with thousands of smallholder farmers across rural Zambia. GNA supports farmers through a structured input-loan and crop buyback model designed to increase productivity, income stability, and market access.

Under this model, farmers receive:

- Input package loans and access to various asset loans
- High-touch Extension services and planting guidance throughout the growing season
- Crop buyback agreements at harvest where the farmers can repay their loans in kind (through crop) or cash and GNA purchases the net quantity of the farmers crop at competitive prices.

For GNA, accurately estimating expected yield per farmer is critical for:

- Improving extension targeting and support allocation, in real time and season over season
- Identifying farmer segments at risk of underperformance and how we can help to improve yields
- Optimizing input package design for maximizing yields and long-term soil health
- Forecasting total crop procurement volumes, directly informing logistics, processing, and sales
- Managing working capital and loan exposure
- Understanding true yields vs how much farmers sell on to GNA, helping to answer how much is held for consumption, recycling seed, or selling to other buyers
- Measuring the success of our input packages versus competitors'

The challenge for students is to use real operational data to generate insights and predictive models that can support better decision-making and risk management.

1. **Predictive Objective**

Develop a model to predict expected farmer yield (or crop buyback potential) using the available data from the provided datasets. Evaluate model performance and identify the most significant predictors.

2. **Input Effectiveness**

Which components of the loan package are most strongly associated with improved yield outcomes? Are there nonlinear or interaction effects?

3. **Risk Identification**

Can early-season data be used to identify farmers at high risk of underperformance? Can farmers be segmented by risk profile and can we flag high-risk farmers early in the season.

Teams should provide empirical analysis, model justification, and actionable recommendations for operational decision-making.

Datasets

GNA uses an in-house built digital system that collects detailed operational data across the farming cycle including the datasets below that are all related to one farming season:

- **farmer_details.csv**- Farmer demographic and geographic characteristics

Field	Description
farmer_id	unique identifier for each farmer
gender	farmer gender
farmer_created_at	date the farmer was registered in the system
dob	farmer date of birth
default_payment_method	default method of payment to farmer during crop buyback
association	farmer type- organic (certified,in-conversion) or non-organic (GNA farmer)
camp_name	farmer designated camp location
district_name	farmer designated district location
region_name	farmer designated region location
number_seasons	number of seasons the farmer has worked with GNA
agroecological_zone	agroecological zone the farmer is located in based on agricultural block/camp classification

- **loan_details.csv**- loan package composition and value

Field	Description
id	table primary key
farmer_id	unique identifier for each farmer
crop_variety_name	crop variety name
crop_class_name	crop class name
initial_down_payment_value	amount expected to be paid as downpayment for each package
package_name	name of the package obtained by the farmer
package_cash_repayment	amount expected as cash repayment for packages obtained on loan
package_in_kind_repayment	quantity (in kgs) expected as crop repayment for packages obtained on loan
package_hectares	number of hectares expected to be planted for each input package
account_package_created_at	date the farmer registered for the package
program_name	type of program the farmers package is enrolled under (farmers can be under multiple programs). Crop programs: Seed, Source, Organic, Partnerships. Non-Crop programs: Assets, Pre harvest loans, Family Package
fertilizer	does the package have fertilizer
fungicide	does the package have fungicide
gypsum	does the package have gypsum
inoculant	does the package have inoculant
insecticide	does the package have insecticide
lime	does the package have lime

seed_guard	does the package have seed guard
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- **buyback_details.csv**- farmer buyback quantities and loan repayment data

Field	Description
id	table primary key
farmer_id	unique identifier for each farmer
crop_class	crop class name
crop_variety	crop variety name
bbk_payment_method	method of payment to farmer for crop bought
base_price	crop base price
grade_a_price	crop grade A price- best grade
grade_b_price	crop grade B price
grade_c_price	crop grade C price- lowest grade
total_weight	total quantity (in kgs) brought by the farmer for loan repayments and GNA purchase (Target variable)
grade_a_weight	total quantity that is grade A
grade_b_weight	total quantity that is grade B
grade_c_weight	total quantity that is grade C
total_net_weight	net quantity of crop that remains to be purchased after deducting crop loan repayment
net_owed_to_farmer	net amount (in Zambian Kwacha) that is owed to the farmer for the net weight purchased by GNA
grade_a_cash_value	cash value of total quantity that is grade A based on grade A price
grade_b_cash_value	cash value of total quantity that is grade B based on grade B price
grade_c_cash_value	cash value of total quantity that is grade C based on grade C price
bbk_created_at	date the farmer brought their crop for GNA buyback

- **planting_survey.csv**- farmer planting information.

Field	Description
id	table primary key
farmer_id	unique identifier for each farmer
crop_planted	crop class planted. N.B- beans could be either sugar beans or navy beans
Qty_kgs_planted	total quantity of seed planted in kgs
planting_date	date that the farmer started planting the seed
spacing_cm_btwn_rows	the spacing in centimeters between planting rows
more_than_one_seed_in_hole	did the farmer plant more than one seed in a planting hole
rcvd_crop_training	did the farmer receive training on how to grow the crop they planted